DECK 11 · WEEK 6 · MORNING 1

Equity I
Markets, Indices & Security Types

How equity trades, how we measure the market, and what we are actually buying

CFA Level I · Vol 5 Learning Modules 1–4

Roadmap — Equity I — Markets, Indices, Efficiency & Security Types

4 sections · ~2 hours · Built bottom-up from CFA readings

01

Market structure & order types

functions of the financial system · market vs limit vs stop · limit order book · quote- vs order-driven vs brokered · call vs continuous

02

Security market indexes

target market & selection · price / equal / cap / float / fundamental weighting · divisor mechanics on stock splits · rebalance vs reconstitute

03

Market efficiency

intrinsic vs market value · weak / semi-strong / strong · anomalies (calendar, size, value, momentum, earnings surprise) · behavioral finance

04

Equity security taxonomy

common vs preferred · callable / putable / cumulative / participating / convertible · public vs private · ADR / GDR / GRS

Learning objectives

By the end of this deck you should be able to:

1. Distinguish market, limit, and stop orders — and trace how a marketable limit buy walks up a live limit order book.

2. Compute the value and return of price-weighted, equal-weighted, market-cap-weighted, and float-adjusted indexes, and adjust the divisor correctly for a stock split.

3. State the three forms of the efficient market hypothesis, identify which form each documented anomaly challenges, and give the standard counter-explanation.

4. Classify an equity security by common vs preferred and by bolt-on features (callable, putable, cumulative, participating, convertible); decompose a convertible preferred into a bond floor plus an embedded call.

5. Compare public vs private equity routes (VC, LBO, PIPE) and describe how an investor accesses non-domestic equity via direct listing, ADRs, GDRs, and GRSs.

SECTION 01

Market structure & order types

Financial systems perform three jobs at once — they move capital, pool risk, and discover prices; orderly trading venues make all three cheaper

Exchanges, OTC venues, and brokered markets all exist to reduce the cost of the same bilateral search: matching someone with cash to someone with an asset.

1 · SAVE & BORROW

Shift purchasing power through time. Individuals save for retirement; firms borrow to fund investment. Both sides need a market to meet.

2 · POOL & DIVERSIFY RISK

Transform lumpy, idiosyncratic exposures into tradeable pieces. Insurance, securitization, derivatives, and listed equity all pool risk across many holders.

3 · DISCOVER PRICES

Aggregate private views into a public price. The best-bid / best-ask spread is the visible output; capital-allocation efficiency is the social benefit.

THE COMMON THREAD

All three jobs get cheaper

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §1 "Functions of the Financial System", pp. 5–12.

Every order is a triplet of instructions — execution, validity, and clearing — and market vs limit is the single most important execution decision

Market order = speed at the cost of price certainty. Limit order = price protection at the cost of fill uncertainty.

Instruction set	What it specifies	Common variants
Execution	HOW to fill — what price and size conditions to accept	Market · Limit · Marketable-limit · AON · Hidden / Iceberg
Validity	WHEN the order can be filled	Day · GTC · IOC (fill-or-kill) · Market-on-close · Stop

MARKET ORDER — TAKES THE MARKET

Executes immediately against the best resting offer. Speed is guaranteed; price concession is not. Dangerous in thin markets — a large market order can walk several levels up the book.

LIMIT ORDER — MAKES THE MARKET

Sets a worst-acceptable price. If unmatched it joins the limit order book and becomes a standing offer. Better price — but may never execute. Aggressively-priced limits (above best offer on buy) are "marketable".

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §12 "Orders and Execution Instructions", pp. 44–50.

A limit order book ranks resting orders by price then time; "the market" is the best bid / best offer, and anything inside it makes a new market

Illustrative book for ticker RENKO. The market is 26 bid, 28 offered. Price-time priority is the dominant matching rule on modern exchanges.

Bid size	Price	Offer size
	33 far	100
	32	60
	31 behind	40
	30	25
	29 near	15
	28 BEST	10
—	27 inside	—
12	26 BEST	
25	25 near	
40	24 behind	
60	23	
100	22 far	

AT / INSIDE / BEHIND

At the market = best bid or best offer. Inside the market = between them (a new limit here makes a new market). Behind / away = beyond the best — may never fill.

AGGRESSIVE VS PASSIVE

Buy 29 is aggressively priced (above best bid); buy 24 is passive. Aggressive orders fill faster but at worse prices — the core liquidity trade-off.

PRIORITY RULES

1. Price priority: highest buy / lowest sell first.
2. Display precedence: visible size beats hidden at same price.
3. Time precedence: first-in wins among equals.

Filling a 15-lot marketable limit buy in a continuous order-driven market clears the book from the best offer upward — a worked trade walk-through

Scenario — the continuous order-driven market for ticker AZURIS has the following book BEFORE our arriving order. We submit: BUY 15 shares, limit 100.5, day.

Bid size	Price (¥)	Offer size
	100.6	12 (Miku)
	100.5	6 (Sakura)
	100.4	4 (Hina)
10	100.3	
8	100.2	
15	100.1	

SOLUTION — WALK THE BOOK UPWARD

Tsubasa submits BUY 15 @ limit 100.5 (day). Discriminatory pricing rule applies.

1. Step 1 — take Hina at 100.4 for 4 lots (best offer). Remaining: 11 lots.
2. Step 2 — take Sakura at 100.5 for 6 lots (still at or below limit). Remaining: 5 lots.
3. Step 3 — Miku is offered at 100.6 > limit 100.5 → STOP. Residual 5 lots rest on the book as a new best bid at 100.5.
4. Avg fill price = (4·100.4 + 6·100.5) / 10 = 100.46. Post-trade book: best bid 100.5 (5 @ Tsubasa), best offer 100.6.

⇒ 10 lots filled at avg ¥100.46; 5 lots standing. Spread tightens from 100.3/100.4 to 100.5/100.6.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, Example 27 "Filling a Large Order", pp. 58–60.

Stop orders and marketable limits are conditionals layered on top of the basic pair — useful for risk management, dangerous as liquidity destroyers

The limit price says WHAT price is acceptable. The stop price says WHEN the order becomes valid. They are orthogonal — and can be combined.

Order type	Trigger condition	Fill rule once valid	Typical use
Market	always valid	take best available	Immediate exit
Limit (buy)	always valid	fill ≤ limit price	Price-sensitive entry
Stop-market (sell)	price trades at/below stop	take best available	Cap downside on long
Stop-limit (sell)	price trades at/below stop	fill ≥ limit price (or wait)	Cap downside, avoid bad prints
Marketable-limit (buy)	always valid	limit set above best offer → immediate partial/full fill	Fast fill, bounded slippage

WHY THE CFA TEXT CALLS STOPS "STOP-LOSS"

A trader long at 40 submits "GTC, stop 30, market sell". If RENKO prints 30, the stop condition is satisfied and a market sell is released into the book — which may execute well below 30 if the market is falling fast. The stop triggers the order but does not guarantee a 30 fill.

[1] Stop-sell orders trigger when prices fall and then add more sell pressure — a positive-feedback loop that can amplify flash crashes.

[2] A stop-limit combines a trigger (stop) with a price floor (limit) — safer than stop-market in thin markets, but may not execute.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §13 "Validity Instructions; Stop Orders", pp. 49–51.

Secondary markets come in three flavours — quote-driven, order-driven, and brokered — and most modern venues blend them

Pick the venue that matches the instrument: dealer markets for capital-heavy inventory; exchanges for standardised instruments; brokers for unique blocks.

Venue type	How trades happen	Dominant asset classes	Price transparency
Quote-driven (dealer)	Customer trades AT the dealer's quoted bid/ask	Bonds, currencies, most spot commodities	Low–mid
Order-driven (exchange)	Electronic order-matching engine pairs buyers & sellers	Listed equities, listed futures, listed FX	High
Brokered	Broker runs a bilateral search; bespoke terms	Real estate, IP, large share blocks, art	Low

BLENDED VENUES ARE NOW THE NORM

NYSE and Nasdaq are order-driven with designated market makers (quote-driven layer). OTC swap markets post electronic dealer quotes (quote-driven with screen transparency). Upstairs block desks hand-cross large equity orders off-exchange (brokered supplement to order-driven core). The three types are ideal-type poles, not mutually exclusive bins.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §15 "Secondary Security Market Structures", pp. 55–60.

Call auctions concentrate liquidity at a single crossing price; continuous trading gives immediacy but at the cost of a wider spread

Most continuous markets open and close with a call auction. Open cross sets the day's opening print; close cross sets the reference price for settlement and NAV.

CALL AUCTION

All orders batched. Single uniform price chosen to maximise volume traded (supply–demand cross). Between calls the market is illiquid — no trades possible.

Used for market opens, market closes, IPO price discovery, and some lower-liquidity venues throughout the day.

Price: uniform. Participation: pooled. Cost: must wait until the call.

CONTINUOUS TRADING

Orders match against resting liquidity any time the market is open. Discriminatory pricing rule: the arriving order pays the limit price of each standing order it consumes (walks the book).

Buyers pay fast, sellers hit the bid fast. Spreads compensate liquidity providers for adverse selection.

Price: path-dependent. Participation: continuous. Cost: the spread.

Source: CFA Institute, Curriculum Vol 5, Learning Module 1, §15 Trading Sessions / Execution Mechanisms, pp. 56–58.

SECTION 02

Security market indexes

Every index is built in three decisions — target market, security selection, and weighting — and each choice silently shapes the reported return

An index is an opinion about what "the market" means — operationalised in rules. Two indexes on the same universe can report very different returns on the same day.

Decision	What it fixes	Example
1 · Target market	Asset class, geography, exchange, sector, cap size, style	S&P 500 = large-cap US equity
2 · Security selection	All constituents vs sample; fixed N vs variable N; rules vs committee	FTSE 100 = top 100 by cap; BSE SENSEX = committee
3 · Weighting	Price / equal / market-cap / float-adjusted / fundamental	Dow = price-weighted; S&P 500 = float-cap

RULES-BASED VS COMMITTEE

Rules-based indexes (FTSE, MSCI, CRSP) re-run a mechanical selection each reconstitution — predictable, backtestable, transparent. Committee-run indexes (S&P 500, BSE SENSEX) allow human override to handle edge cases (mergers, distress, sector balance) — at the cost of some opacity.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §2 "Index Construction", pp. 82–88.

The single-period return of an index is just a weighted average of its constituents' returns — price return vs total return differ only by the dividend leg

BOP weights determine each stock's contribution; EOP weights show where the index drifted to.

$$R_index = \Sigma_i \; w_i_BOP \times R_i \qquad R_total = R_price + (dividends / BOP\ index\ value)$$

where: w_i_BOP = constituent weight at beginning of period; R_i = single-period constituent return.

MINI-EXAMPLE — 2-STOCK PRICE-WEIGHTED INDEX

HELION BOP 50, EOP 55, dividend 0.75. RENKO BOP 25, EOP 22, dividend 0.10. Divisor = 2.

1. BOP index = (50+25)/2 = 37.50. EOP index = (55+22)/2 = 38.50.
2. Price return = (38.50−37.50)/37.50 = +2.67%.
3. Total dividends paid on 1-share-each basis = 0.85. Total value flow = price change 1.00 + div 0.85 = 1.85.
4. Total return = 1.85/37.50 = +4.93%.

Price return 2.67% vs total return 4.93% — dividends added 226 bps in this single period.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §2 "Calculation of Single-Period Returns", pp. 83–85.

Price weighting makes high-price stocks dominate by accident of quotation — and a stock split forces the divisor to be re-scaled to preserve index level

Price-weighted weights reflect only the share price — not market cap, not economic size. A $200 stock carries 10× the weight of a $20 stock regardless of how big either company is.

$$w_i_P = P_i \;/\; \Sigma_j P_j \qquad \text{Index value} = \Sigma P_i \;/\; \text{Divisor}$$

Security	BOP price	BOP weight	EOP price	Price return	EOP weight
HELION	50.00	49.26%	55.00	+10.00%	52.38%
RENKO	25.00	24.63%	22.00	−12.00%	20.95%
AZURIS	12.50	12.32%	8.00	−36.00%	7.62%
NOVEX	10.00	9.85%	14.00	+40.00%	13.33%
CORYN	4.00	3.94%	6.00	+50.00%	5.72%
Sum / Index (divisor 5)	**101.50 → 20.30**	**100.00%**	**105.00 → 21.00**	**+3.45%**	**100.00%**

THE HEADLINE INSIGHT

HELION drives 49% of the i

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §3 "Price Weighting", Exhibits 1–2, pp. 87–89.

Comparing price-weighted vs market-cap-weighted after a 2-for-1 split in HELION — same five stocks, two different "market" readings

Scenario — at EOP, HELION does a 2-for-1 split. Share price 55 → 27.50, shares 3,000 → 6,000. Market cap unchanged. But the price-weighted index must rescale its divisor.

WORKED EXAMPLE — SPLIT MECHANICS

Compute both indexes BEFORE and AFTER the HELION 2-for-1 split. Show why the price-weighted divisor changes and the cap-weighted index does not.

1. Price-weighted BEFORE split: Σ prices = 55+22+8+14+6 = 105. Divisor = 5. Index = 21.00.
2. After split, ΣP = 27.50+22+8+14+6 = 77.50. To keep index at 21.00: new divisor = 77.50 / 21.00 = 3.6905.
3. Cap-weighted BEFORE: ΣMCap = 3000·55 + 10000·22 + 5000·8 + 8000·14 + 7000·6 = 165k+220k+40k+112k+42k = 579k.
4. Cap-weighted AFTER split: HELION shares 3000→6000 at price 55→27.50 → mcap unchanged at 165k. Total unchanged at 579k. Index value unchanged.

⇒ Price-weighted: divisor 5 → 3.6905, index preserved at 21.00. Cap-weighted: no adjustment — split is a non-event.

Security	Price post-split	Weight P-W after	Mkt cap (k)	Weight cap-W
HELION	27.50	35.48%	165.0	28.50%
RENKO	22.00	28.39%	220.0	38.00%
AZURIS	8.00	10.32%	40.0	6.91%
NOVEX	14.00	18.07%	112.0	19.34%
CORYN	6.00	7.74%	42.0	7.25%

Equal, market-cap, float-adjusted, and fundamental weightings each solve one problem and create another — the choice is always a trade-off

Float-adjusted cap-weighting is now the industry default for broad indexes — combines investability, minimal rebalancing, and representativeness.

Method	Weight rule	Advantage	Disadvantage
Price-weighted	$w_i = P_i / \Sigma P$	Dead-simple computation	Arbitrary — splits redistribute weights
Equal-weighted	$w_i = 1/N$	Small-cap / size-effect tilt free	Constant rebalancing; underweights big names
Market-cap	$w_i = (Q_i P_i) / \Sigma(Q_j P_j)$	Self-rebalancing; investable	Concentrates in overvalued names (momentum drift)
Float-adjusted cap	Cap × % publicly tradable	Excludes locked-up / strategic holdings	Float data must be maintained by provider
Fundamental	$w_i \propto$ sales, book, CF, or dividends	Decouples from price —	Opinionated — introduces value tilt

KEY INTUITION

Cap-weighted indexes are "the market" under CAPM equilibrium (every investor holds the market portfolio in proportion to supply). Fundamental-weighted indexes are an active bet against the market — even if dressed up as passive.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §3 Weighting Methods, pp. 89–95.

Indexes need rebalancing (weights drift) and reconstitution (constituents drop in/out) — two different kinds of turnover that every passive fund must track

Cap-weighted indexes barely need rebalancing; equal-weighted indexes need constant rebalancing. Both need periodic reconstitution.

REBALANCING — REFRESH WEIGHTS

Reset weights back to the target rule. Cap-weighted: mostly automatic (weights drift to their "true" value when prices move). Equal-weighted: requires selling winners and buying losers every rebalance — meaningful turnover, tax drag, trading costs.

Frequency: monthly, quarterly, or semi-annually for most major indexes.

RECONSTITUTION — REFRESH MEMBERSHIP

Add and remove constituents. Driven by target market rules — e.g., a stock that drops below size / liquidity thresholds exits. Additions often pop on announcement (index-inclusion effect) as passive funds pre-position.

For cap-weighted indexes, reconstitution forces a weight recalculation of ALL names — not just the ones that moved.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §4 "Index Management", pp. 95–97.

Equity indexes are organised into four families — broad, multi-market, sector, and style — each serving a different analytical question

Pick the index that matches the question. "Did US large caps rise?" → S&P 500. "Did tech outperform staples?" → sector indexes. "Did value beat growth?" → style indexes.

Family	What it represents	Typical uses	Examples
Broad market	Large slice of a domestic equity market	Market sentiment · benchmark · asset allocation	S&P 500, FTSE All-Share, TOPIX
Multi-market	Bundle of countries or regions	Global allocation · EM vs DM tilt	MSCI EAFE, MSCI ACWI, FTSE Emerging
Sector	Companies in a single industry group	Sector rotation · industry analysis	S&P 500 Financials, MSCI World Energy
Style	Size × value/growth classification	Style drift · factor exposure benchmark	Russell 1000 Growth, Russell 2000 Value

THE 3×2 STYLE BOX

Size dimension: large / mid / small cap. Style dimension: value / growth. Gives six canonical style categories (e.g., "large-cap value", "small-cap growth") that anchor manager benchmarking and factor attribution.

Source: CFA Institute, Curriculum Vol 5, Learning Module 2, §6 "Equity Indexes", pp. 99–102.

SECTION 03

Market efficiency

Market efficiency says prices adjust quickly and accurately to new information — intrinsic value can still differ from market value, but only briefly

Efficiency is a spectrum — driven by how many informed participants trade, how fast information spreads, how costly it is to trade, and what limits-to-arbitrage are present.

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Market value ≈ Intrinsic value      Abnormal return = Actual return - Expected (risk-adjusted) return
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where: In a fully efficient market expected abnormal returns on any trading rule are zero — after transaction costs and risk adjustment.

Factor that pushes TOWARD efficiency	Factor that pushes AWAY from efficiency
Large number of profit-motivated informed traders	High information-acquisition costs
Mandatory timely disclosure (10-K, IFRS quarterly)	Transaction costs and bid-ask spreads
Low trading costs + deep liquidity	Short-selling restrictions / stock borrow costs
Well-functioning derivatives & arbitrage tools	Limited investor attention / behavioral biases

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §2 "Concept of Market Efficiency", pp. 125–131.

Three forms of EMH stack like Russian dolls — weak ⊂semi-strong ⊂strong — and the evidence supports only the first two in developed markets

Each form is a statement about what information is already priced — and therefore what trading strategy cannot earn abnormal returns.

Form	What prices reflect	Useless trading strategy	Developed-market evidence
Weak	All PAST market data (prices & volume)	Technical analysis / chart patterns	Largely supported
Semi-strong	All PUBLIC information (weak + fundamentals)	Fundamental analysis based on public data	Broadly supported
Strong	All PUBLIC + PRIVATE information	Insider trading	Rejected — insiders do profit

IMPLICATION — WHY ACTIVE MANAGEMENT STRUGGLES

If markets are semi-strong efficient, skill in analysing public information does not reliably generate alpha — the information is already priced. Mutual-fund studies bear this out: net-of-fees, the average active manager underperforms the benchmark. Fundamental analysis still has value: it is the mechanism by which markets ACHIEVE semi-strong efficiency.

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §4 "Forms of Market Efficiency", pp. 132–137.

Documented anomalies span calendar, cross-sectional, and event-driven effects — but most shrink or vanish once risk-adjusted and net of transaction costs

Rule of thumb — an anomaly is real only if it survives three tests: out-of-sample replication, risk adjustment, and after-transaction-cost profitability.

Anomaly	Category	Observation	Standard counter-explanation
January effect	Time-series	Small-cap returns spike in early January	Tax-loss selling + window dressing — faded post-2000
Momentum (3–12m)	Time-series	Past winners keep winning over 3–12m horizon	Rational shocks to cash-flow growth (Johnson 2002)
Size effect	Cross-sectional	Small caps outperform large caps risk-adjusted	Present only in microcaps; illiquidity premium
Value effect	Cross-sectional	Low P/B stocks beat high P/B stocks long-run	Fama–French 3-factor: distress / value risk factor
Earnings surprise drift	Event-driven	Positive surprises drift up AFTER announcement	Survives in studies; weakens after transaction costs
Closed-end-fund discount	Event-driven	CEFs trade 4–10% below NAV	Illiquidity + deferred-tax-liability overhang
IPO long-run underperformance	Event-driven	IPOs beat day-1, then underperform over 3–5y	Methodology artifact + initial overreaction

Behavioral finance offers a microfoundation for anomalies — loss aversion, herding, and overconfidence — but the market can still clear efficiently even if individuals do not

Efficiency does not require that every investor be rational — only that enough rational arbitrageurs force prices back to fundamentals.

LOSS AVERSION

Pain of a loss outweighs pleasure of an equal-size gain (~2× in experiments).

Consequence: disposition effect — investors hold losers too long, sell winners too early.

Can explain overreaction to bad news — but so can undereaction, so behavioral theory alone is not decisive.

HERDING & INFORMATION CASCADES

Investors trade on the same side of the market simultaneously — often ignoring private information.

Can create momentum (herd toward rising names) or overreaction (herd away from distressed names).

Rational micro-foundation: information cascades — each investor rationally infers others know something they do not.

OVERCONFIDENCE

Investors overestimate precision of their own forecasts — especially after a recent success.

Leads to excessive trading, under-diversification, and mispricing of low-probability tail events.

In aggregate may produce return volatility in excess of fundamentals — but arbitrageurs still force mean reversion.

Source: CFA Institute, Curriculum Vol 5, Learning Module 3, §8 "Behavioral Finance", pp. 146–149.

SECTION 04

Equity security taxonomy

Common shares give ownership, residual claim, and voting rights; preferred shares rank higher on cash flows but typically vote only when dividends are in arrears

Preferred shares are a hybrid — bond-like cash flows, equity-like perpetual life, second-to-last in liquidation.

Feature	Common shares	Preferred shares
Dividend	Discretionary; can be zero	Fixed rate (stated %); not a contractual obligation
Dividend priority	Last — after all preferred and debt	Before common; may accumulate if cumulative
Liquidation priority	Last (residual claimant)	Before common, after debt
Voting rights	One vote per share (typical); statutory / cumulative	None typically; voting triggered if divs in arrears
Participation in earnings growth	Full — captures upside via price & dividend growth	Limited — cash flows capped at stated coupon unless participating
Maturity	Perpetual	Usually perpetual; may be callable / putable

CUMULATIVE VS STATUTORY VOTING FOR COMMON SHARES

Statutory = one vote per share per candidate (one-share-one-vote per line). Cumulative = total votes can be concentrated on a single candidate — protects minority shareholders by making board representation achievable below 50% ownership.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 Common + Preference Shares, pp. 165–170.

Bolt-on features — callable, putable, cumulative, participating, convertible — turn a plain preferred share into five economically different securities

Each bolt-on is an embedded option that shifts value between issuer and investor. The shareholder's return is the coupon plus the net value of all embedded options.

Feature	Who holds the option	What it does	Direction of value
Callable	Issuer	Issuer can buy back at set price — valuable if rates fall	Negative to investor
Putable	Investor	Investor can sell back to issuer at set price	Positive to investor
Cumulative	Investor (automatic)	Missed dividends accrue and must be paid before common divs	Positive to investor
Participating	Investor (automatic)	Extra dividend when profits exceed threshold; upside in liq.	Positive to investor
Convertible	Investor	Can swap into a fixed number of common	Positive to

WHY VC DEALS ALMOST ALWAYS USE CONVERTIBLE PREFERRED

Convertible preferred = bond-like floor (safety if the startup fails) + call option on common (full upside if the startup succeeds). Matches VC payoff asymmetry better than either straight preferred or straight common. Combined with cumulative + participating features, it also protects investors in interim liquidations.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 Preference Shares, Exhibits 9–10, pp. 169–170.

A convertible preferred decomposes into a fixed-income floor plus an embedded call option on the common — a two-component valuation

Price floor = value as a straight preferred. Conversion value = today's value of the common shares it can convert into. Traded price ≥ max(floor, conversion value).

P_convertible = max(Bond floor , Conversion value) + Time value of the embedded option

where: Bond floor = PV of coupons & par as straight preferred · Conversion value = conversion ratio × common share price

AZURIS SERIES C 6% CONVERTIBLE PREFERRED — DECOMPOSITION

Par USD 50 · annual coupon 6% (USD 3) · conversion ratio 2.5 · AZURIS common trading at USD 18 · market yield on comparable straight preferreds = 7.14%. Observed market price of the convertible preferred = USD 48.

1. Bond floor: PV of a perpetual 3/yr cash flow at 7.14% ≈ 3 / 0.0714 = USD 42.00.
2. Conversion value: 2.5 × USD 18 = USD 45.00.
3. Floor-or-convert: max(42.00, 45.00) = USD 45.00 (would convert today to capture the common).
4. Market price 48.00 − 45.00 = USD 3.00 = time value of the remaining optionality (waiting to convert at a higher common price).

⇒ P = 45 (intrinsic) + 3 (option time value) = USD 48.00. Above the bond floor by 6.00 — equity-like; below pure common of 45, but with downside protection at 42.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §2 Convertible Preference Shares, pp. 169–170.

Private equity — VC, LBO, PIPE — and public equity occupy the same asset class but with wildly different liquidity, disclosure, and return patterns

The premium on private equity is partly compensation for illiquidity and partly compensation for the active-management / governance work performed by the GP.

VENTURE CAPITAL

Seed / early / mezzanine capital into pre-profit companies.

Instrument: convertible preferred (cumulative, participating, anti-dilution).

Hold: 5–10y. Exit: IPO or trade sale. Failure rate is high; winners subsidise losers via portfolio construction.

LEVERAGED BUYOUT (LBO / MBO)

Investor group uses heavy debt to take an already-public company private.

Targets: undervalued asset-heavy firms with steady cash flow (to service debt).

Value creation: operational improvement + financial engineering + re-listing at higher multiple years later.

PRIVATE INVESTMENT IN PUBLIC EQUITY (PIPE)

Public company sells sizeable equity stake to a private investor(s), usually at discount to market.

Used when the issuer needs capital fast and wants to avoid a long registered secondary offering.

Hybrid: public liquidity for the underlying shares, private-negotiation economics on the block.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §3 "Private Versus Public Equity Securities", pp. 170–173.

Investors reach foreign equity via direct listing, depository receipts (ADR / GDR), or global registered shares — each choice trades FX friction for regulatory burden

The depository bank holds the underlying foreign shares and issues receipts denominated in the investor's currency — turning a cross-border trade into a local one.

Route	Trades on	Currency	Regulatory overhead for issuer	Typical investor
Direct investing	Issuer's home exchange	Foreign	None extra (domestic listing only)	Institutional
ADR Level I (unlisted)	US OTC pink sheets	USD	Minimal — Form F-6 only	Retail + inst.
ADR Level II (listed)	NYSE / NASDAQ / AMEX	USD	Full SEC disclosure, no new capital	Broad US market
ADR Level III (listed, capital)	NYSE / NASDAQ / AMEX	USD	Full SEC + public offering	Broad US market
ADR Rule 144A	Private QIB placement	USD	None with SEC (Rule 144A carve-out)	QIBs only
GDR	London / Luxembourg typically	Usually USD	Local disclosure, no SEC	Non-US inst.
Global Registered Share (GRS)	Multiple exchanges simultaneously	Local of each exchange	Highest — multi-jurisdiction filings	Global inst.

Source: CFA Institute, Curriculum Vol 5, Learning Module 4, §4 "Non-Domestic Equity Securities", pp. 173–178.

KEY TAKEAWAYS

Deck 11 — Equity I — Markets, Indices, Efficiency & Security Types

- Every order is execution × validity × clearing; market orders TAKE liquidity (speed, price risk), limit orders MAKE liquidity (price protection, fill risk).
- A limit order book ranks by price then time; arriving marketable orders walk the book, paying each standing order's limit price under the discriminatory pricing rule.
- Venues blend three archetypes — quote-driven (bonds, FX), order-driven (listed equities), brokered (unique blocks). Call auctions vs continuous trading differ by liquidity concentration.
- Indexes are defined by target market × security selection × weighting. Price weighting is dominated by high-priced names and needs divisor rescaling on splits; cap weighting is self-rebalancing; equal weighting requires constant turnover.
- EMH: weak ⊂ semi-strong ⊂ strong. Developed markets broadly support weak & semi-strong forms. Insider profits reject the strong form. Most documented anomalies shrink once risk-adjusted and net of costs.
- Behavioral finance (loss aversion, herding, overconfidence) gives microfoundations to anomalies — but market efficiency only requires enough arbitrageurs, not rational individuals.
- Preferred shares = bond-like priority + equity-like perpetual life. Five bolt-ons: callable (issuer), putable (investor), cumulative, participating, convertible. Convertible preferred = bond floor + embedded call on common.
- Non-domestic access: direct, ADR I–III + Rule 144A, GDR, GRS. Trade FX + foreign-rule friction for SEC / multi-jurisdiction disclosure friction.